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H8SCOOP

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BITS & PIECES

NEW H8SCOOP PHONE HOURS

I am now working on computer support and H8SCOOP on a full time basis. I wanted to make you aware of this, since it will change the hours you are likely to catch me by phone. You can feel free to call about any time and take your chances. I try to stick around here most of the time. When I have to step out, you will either get my wife who will take messages and orders, or no answer in which case you can call back. The only stipulation I make, is I will accept NO PHONE CALLS between sunset Friday evening and sunset Saturday evening, as that is the Sabbath, my day of rest and worship.

MORE ON 400K DRIVES

Just a few more notes on the 400K drive package I am offering. I will hold my price of \$495 for the bare drive with software till the middle of August, then the price goes up to \$525 for the drive and software package, or \$500 for bare drive. Now the bare drive alone is \$470. The software is available through HUG, but I prefer you purchase it through H8SCOOP, as the author, Dean Gibson gets a 50% royalty through H8SCOOP, and nothing when HUG sells it, they just pocket the whole thing (good deal for HUG)! My disc also contains more documentation to help you get your system up with less trouble and time.

Questions have popped up on the disc media and I want to clarify this. The majority of your standard discs should work. I have been using Memorex single side single density and have had about 99% success. The drive package still uses 10 hole hard sector discs, and if you have any decent type of disc such as VERBATIM or MEMOREX, you should have no problems.

As far as warranty, the drives are covered parts and labor for 90 days, any problems send them to me and I will take care of it.

Since it is very difficult to obtain cabinets and power supplies for the drives, I decided to handle them. The price is not the best, but at least you can get them. I will sell single cabinet with power supply (commercial quality) for \$100 with the drive purchase, or \$110 seperately. It is a very compact and nice looking unit.

I will also be distributing the H89 double density controller board (complete with cables for 5 1/4" drives, and 8" drives, plus CP/M 2.2 needed to operate these drives. My price on this will be \$510. Latter this month, the software should be available for this double density card under HDOS, which means 800K per Tandon drive, and 200K per standard drive. I'll keep you posted.

FEEDBACK ON REMARKABLE EXPERIENCE

SSgt James Romoser sends the following info: "You probably know that Hoyle & Hoyle Software is offering an error free update version of 'A Remarkable Experience'. To get the updated version, send \$1.00 and the original diskette. Companies like Hoyle & Hoyle Software should be mentioned in your 'Good Guy' column and given our support. We need more like them."

From Hoyle & Hoyle "We are receiving many requests for help getting into "A Remarkable Experience" cave, and thought you might like to offer your readers some basic hints."

1) There are actually five separate ways to enter the cave. 2) Only two of these have much likelihood of being discovered in the beginning. 3) You will notice that when you try to go North from the pool that you are told, the pool is in the way instead of there is no way to go that direction. Try swimming. 4) There will be a boulder in the way. What would you do if you came upon a bolder on the path and it was too big to move? 5) Now to the passwords for entrance. There are three passwords that you must discover. Some may have more than one part. To get in the cave, say IN, ENTER, DOOR, NORTH, etc. Then when asked for the password, say the three words one at a time. I'll tell you the first one.

Einstein was a very _____ scientist, but not us. You fill in the blank with the word FAMOUS and take away the word US and you are left with the first password, FAME. Happy adventuring! See NEW STUFF for more from Hoyle & HOYLE.

ANSWER TO REQUEST ON 255 VOLUME NUMBER LIMIT

Burt Hulland writes in response to a request in H8SCOOP on the limit of 255 for disc volume numbers. "No, there is NO possibility of increasing this (at least with 5" disks, and I suspect the same is true with 8"). The operating system uses this number to check that the disk has not been replaced without a MOUNT or RESET command (which would be a sure way of destroying the files on a disk if you could do it!). That is why you should try to avoid having two disks with the same volume number - you are defeating one of the safeguards in the system, and will regret it sooner or later. (I have found this out--ed.) That is also one of the problems with CPM; it has no completely effective mechanism for preventing such accidents. In HDOS the volume number is recorded in the header block for each sector (except the first 10 on the disk). Unfortunately, the disk format only allows one byte for that number - thus the restriction of 000 to 255 on the volume number. Any change which would increase this range would make the system incompatible with existing disks."

ON H89 CASSETTE TO SERIAL REQUEST

Again from Burt Hulland "Conversion of the H88 cassette cassette interface to an RS-232 port should be an easy task. I don't have a schematic of it (I'm going to get my hands on one soon), but I have done the same thing to an H8-5 to make a two-port

serial board out of it. Essentially what you do is remove the FSK modulator and decoder circuits (I just pulled the chips), add the RS-232 level converters and connect the 8251's receive clock pin to the transmit clock. You may be restricted to 1200 baud (or 300 baud if you give the UART a mode instruction of 117Q) since there is probably no other clock frequency on the H88/H89 board. For device drivers, you can use Heath's ATH85 or my Epson driver or modify one of the other Heath drivers by replacing the 8250 initialization code and status/data I/O with the 8251 equivalents."

Craig Todtchinder of The National Software Register tells me to get in touch with Al Hufnagel, 832 Balfour Rd, Montgomery, AL 36117, phone (205) 277-3734. He has a conversion that costs 50 cents for the documentation. I thank you for those responses to the REQUEST column. Without the effort of these individuals to respond to requests and general information, we would all be in big trouble. If you have answers, please take the time to respond--you may be on the other end some day!

ON MEMORY CHIPS FROM 16K BOARD

Burt Hulland says he has found you can replace the memory chips on the 16K board with those TI 4044 chips used on the 8K boards, which are readily available. You probably wouldn't want to use too many since they draw more power and may load the address lines more heavily.

Henry Beechhold also writes "Future Electronics may be a source for the WH16 static RAMs. Better, Substitute 2 8K boards. Incidentally, for those using 8K boards, 4041's will sub for 4044's. The 4041 is more tightly spec'd (i.e., its operating parameters are somewhat narrower) than the 4044, but it's cheaper than the 4044 and works exactly the same. Future sells these. I don't have their address at hand, but I think they're in Framingham, MA.

TRIONYX MEMORY UPGRADE

From Myron Seibold taken from MNET, the M-H8 64K dynamic memory board requires the following modification to operate properly with zero based CP/M in the Heathkit H8 computer. This change is also a design improvement which should be incorporated on all TRIONYX M-H8 memory boards.

- 1) Cut Trace to U42-pin 9, solder-side of board.
- 2) Raise pin 11 of U42. 3) Connect a jumper wire from U42-pin 9 to U43-pin 2. 4) Connect a jumper wire from U42-raised pin 11 to U50-pin 8.

ON DOUBLE DENSITY DISK STORAGE FOR H8

There has been much work in double density disk controller systems for the H89, one by CDR SYSTEMS. Thus far, little has been done for the H8. I assume that is because the 89 sales are picking up, while the H8 is decreasing. CDR plans to soon release a double density controller for the H8, as well as issuing a version of software under HDOS (now presently only available under CP/M for H89). I'll keep you in touch with developments as given to me by Capital Heath User's Group. If you want more info on this or the double density board for the H89 which normally goes for \$695 including CP/M, contact CHUG. They are projecting a price of \$432 through CHUG. If you are interested in this for

the H89 or H8, contact CHUG, POB 341, Fairfax, VA 22030.

H8 vs H89?

Henry Beechhold writes "Henry, I hope that you're not going to let the H89 swamp H8! With the mods I have in my system, I feel that I have a computer far superior to the H89, and I look forward to reading as much as I can get about goings on in the world of H8! Of course, with my Z80 board, I can run all the H89 programs, but H89 hardware doesn't have much to offer to H8ers."

My response: I started out supporting H8 only, and then began receiving many requests for H89 support since 1) many H8ers are going to H89's for dual systems and backup systems, and 2) Software wise, and some hardware, such as best memory chip prices, disc prices, etc. are compatible with both. It was with this thought I merged with the H89, and even purchased a few for myself.

I will never sell out the H8, since I strongly believe in them myself (I have 2 of them). For some applications, the H8 is better, mainly for the experimenter and those wishing expansion, while the H89 is better for a compact take it with you, or business system. The problem of late in leaning toward the H89, is that's what's coming in! I can only print what I receive. There has been a tremendous amount of work with the H89 of late, especially with disk controllers. There are several double density controllers for the H89, and even hard disc controllers, and thus far, nothing for the H8. This I have no control over. One company strongly supporting the H8, is Trionyx (see NEW STUFF) and I feel lot of good stuff is coming from them. There was also things for the H8--such as Ted Benglen's Talking H8, and Don Longs 12 port parallel/clock card--The H8 is by no means dead. I will continue to support the H8 to the best of my ability, but if the majority of contributions are from H89'ers, that's what I print. H8ers--Get on the ball and send me your goodies to publish!

ON PAPER TIGER 560 PRINTER DEVICE DRIVER

Chuck Ballinger, RFD 1, Box 676, Spokane, WA 99204 has modified the LP24 device driver to handle all printer output for the Paper Tiger 560 printer. He has the source code as well as documentation, and two MBASIC programs to demo the printing and graphics available. Anyone can have these at no cost as long as you send Chuck a disk, which he would appreciate with some programs on it in exchange, and return postage. You can also contact Chuck on MNET 70235,313.

FEEDBACK ON PILOT MODS (12/80 H8SCOOP)

John Bennin writes to Michael Ryan "I made the modifications to PILOT that you had suggested in the December 1980 issue of H8SCOOP. Up until the present, I did not have much occasion to make use of the language. Today, however, when I made more extensive use of the language, I found that the Compute (C:) function does not work in the modified version. It works as it should in the unmodified version.

After trying a few combinations of the suggested modifications, I found one that seems to work. All of the changes that were suggested were made

except the first two and this seems to work properly. I have not tried all of the PILOT commands with the change, however, and I may have something else screwed up. I would appreciate any comments that you may have on the problem.

For clarification, the changes which caused the problem are:"

052.035 from 376 to 257--changed back to 376

052.036 from 101 to 311--changed back to 101

Response from Michael Ryan:

"I looked into your complaint concerning my suggested modification to 'PILOT'. You are absolutely correct, the changes at locations 052.035 and 052.036 do cause a problem when using the (C:) command. When I originally modified the program, I looked for any routine that checked or limited input to upper case only. It is apparant that the (C:) command requires the input of an uppercase letter for a variable and thus the program must check for this.

"I intended to use 'PILOT' for training my kids to use the computer, however, I got quite busy soon after I modified the program and never really used it. I believe that one of the benefits of Henry's newsletter is that it allows us to pass on information like this to each other. . ."

NEW STUFF

As you can see, this column is taking up a large portion of H8SCOOP this month, and that seems to be the general trend. This is an excellent sign, since it means support of the H8 and H89 is growing at an exponential rate. These systems are becoming more and more popular all the time.

NEW NEWS FROM HEATH CATALOG

I just received my Fall 1981 Heathkit catalog and there are some interesting things within. Note now that the H89 is available with non-glare green screen (the way the Z89's have been for quite some time) for \$1325 compared to \$1295 for standard white CRT (see page 70). Also the H19 is also now available with the green screen for \$725 verses \$695 for standard white. Both are also available with Anti-Glare white for the same prices as the green. I feel this is a very good deal.

On page 72 you'll find the Z-80 CPU for the H8 announced at \$199, which is not too bad, considering it comes with the O-org option on the board. Also on page 74 you'll find a conversion kit for \$695 to convert your H19 to and H88. There is some new software available also, and a 50% reduction in the price of AUTOSCRIBE (ZENITH ELECTRONIC TYPING) from \$395 (which was way too high) to \$195 which is a worthwhile deal.

PAGED EDITOR AVAILABLE UNDER CP/M

Scott Witt, 79 Old Haverstraw Rd, Congers, NY 10920 now has available his PAGED text editor to run under CP/M Org-0 for \$25. It contains all the features of the HDOS version, and more. Scott is also making the HDOS version available for those interested in it for \$25, or both versions for \$45. See the March issue of REMark for discussion of its features.

NEW FROM TARCO

MAXTEK announces a unique computer graphics conversion unit, the XCEL. This hardware allows the H/Z89 to display 16K pixel graphics with 512 X 240 line resolution. The unit can be installed in less than one half hour and allows full memory capability of the H89 to remain intact. Graphic video is independently controlled and overlays normal processor video. The user may select normal video, graphics video, or both; to be displayed via a three way switch. There are five XCEL software packages available for use with the graphics unit; the Symbol Generator, Graph Plotter, 3D generator, Surface plotter and Graphics Terminal Emulator. These packages provide for easy user interface to assembler or high level languages.

This is being distributed by Tarco Development Corp, 1445 Koll Circle, Suite 111, San Jose, CA 95112 (408) 947-1101. Prices? Sit down. Graphics board goes for \$850, Symbol generator-\$170, Graph Plotter \$170, 3D-\$335, Surface Plotter \$380, and Emulator for \$380. Write TARCO for more info.

FROM TRIONYX ELECTRONICS

Trionyx Electronics, POB 5131, Santa Ana, CA 92704 is publishing a "Quick Reference Summary" guide for HDOS 2.0 and CP/M 2.2. These manuals were written by Robert Sutton and should prove to be an invaluable reference guide whenever using the H8 computer. They are intended to be used by those already familiar with the system who need to occasionally refresh their memory regarding procedural details while operating the system. Considerable software available for use with HDOS and CP/M is covered in addition to the operating systems, themselves.

Trionyx recommends these two manuals very highly. Each consists of about 40 double sided pages and are available for \$15 each, or \$25 for both postpaid.

I ordered and installed their new motherboard, the T-H90. For \$125, I received the board with an extra capacitor and the necessary gold pins to bring it to Heathkit standards. The board is very well designed with extra grounding. Before I installed this board, my computer would usually show 277.377 on boot, instead of the 377.377 it should have. I experienced various intermittants, which have been getting worse, especially since I installed the O-org mod to my H8. Since this new motherboard was installed, I booted perfect at 377.377 every time, with no glitches, failures, intermittants or anything.

I feel this board is very well worth the price, and recommend it to EVERYONE using the H8 for any serious computing purposes. The board also has additional positions for small cards like Gregg Saville's sound board, and the O-org board, which do not take any of the additional card positions away. In addition to this, it has 40 pins extra which will allow the computer buss to expand in the future, for Trionyx's latter releases mentioned below.

MASS MEMORY for the H8/H89 is a completely solid state mass storage unit having no moving parts, low power dissipation, silent operation, no

maintenance required, high speed access and expandable from 128K bytes to 5 megabytes. The base price to be under \$500, available latter this year.

Their Z80 CPU which should be released in September this year has a 2 or 4 Mhz clock, calendar clock with Nicad battery and charging circuit, Arithmetic chip, Random number generator, and Front End Buss Termination network. The bare PC board will go for \$75. I'll have more on this one next month.

Projected plans from Trionyx include Voice synthesis, High resolution color graphics board, 16 bit CPU board, and Radio controlled Robot, to mention a few.

THE NATIONAL SOFTWARE REGISTER

The National Software Register, 423 Bedford Rd, Schenectady, NY 12308 is a monthly publication of software for sale by individual programmers, vendors and compaines all across the country. The purpose of the REGISTER is to provide and maintain the largest selection of microcomputer software for all systems listed, in a single publication. "We will bring you new software releases as fast as we get them, if received by the first of the months they will usually appear in that months issue. We are providing the lowest cost advertising in the industry that will allow the part time programmer and even the smallest entrepreneur to compete with the corporate giants by not having to pay the big bucks for advertising. Users are urged to send their requests for software they need and can't find to be listed under WANTED. User wand ads are free to subscribers. Systems served by the REGISTER are: Apple, Atari, Commodore, Compucolor, CP/M, Exidy, Heath, Microsoft Basic, Northstar, Ohio Scientific, Radio Shack, and SWTP. Subscriptions are \$9 for 1 year (12 issues) USA only. Advertising rates: \$.05 per word per issue or \$.40 per word per year (12 issues). Classified ads only. No display adds are accepted.

NEW FROM ACT--5 1/4" HARD DISK CONTROLLER CARDS

From ACT (American Computer and Telecommunications Corp) 11301 Sunset Hills Rd, Reston, VA 22090 (703) 471-9750 comes 5 1/4" controller cards for the H89. These are available off the shelf for \$1,250. This includes CP/M 2.2 software to work with the controller. Simply install the hardware, boot, and away you go with 5 Megs. You can also order fully assembled CRT terminals with an integrated 5 1/4" Winchester Hard Disk and controller card package. Contact ACT for more info. It's available NOW!

ON MX-80 PRINTERS

Those having the MX-80's can order a friction feed from Orange Micro Inc, 3150 E. La Palma, Suite I, Anaheim, CA 92806 (800) 854-8275 for \$75. The friction feed will accept single sheets of your letterhead or multiple copy forms with up to 4 part carbon. The pin feed replaces the adjustable sprocket mechanism allowing the use of 9 1/2" wide continuous fanfold paper which is an industry standard size. No drilling required, takes about 15 minutes to install. Phone orders on MC or VISA are welcomed.

Those interested in good prices on the EPSON MX-80 printer with serial interface, contact Taylor Electronics, PO Drawer 11N, Milwaukee, WI 53201, or phone (414) 241-4321. For about \$460 you can get the printer, and another \$75 gets you the serial interface and cables.

Another device driver is available for the MX-80 (besides the one from HULLAND ENGINEERING) from Lindley Systems, 21 Hancock St., Bedford, MA 01730. Write to them about prices and information.

MORE FROM SKIP BARRON OF MAKO DATA PRODUCTS

Skip Barron of Mako Data Products, 1441-#B N. Red Gum, Anaheim, CA 92806 (714) 632-8583 has available more goodies. The PSG TIMES TWO for the H89 has two AY-3-8910 PSG (programmable Sound Generator chips) to allow H89 owners to enjoy the same sound capabilities as some S-100 machines. Two separate voices, three-note chords for each, full envelope control, 12 octave range, multiple complex sound effects, and now FOUR general purpose parallel I/O ports! This board occupies a slot on the right side of your machine, and runs at the old Cassette Port address, but may optionally be configured for any decoded port. This board is \$125-assembled only, and orders are being accepted for delivery now.

For the H8, the PSG TIMES FOUR. This board has FOUR PSG chips allowing for 4 seperate instrument voices, with chords, or multiple complex sound effects. The EIGHT general purpose, 8 bit I/O ports (64 lines) may be used for game paddles, sense switches, keyboard interface, etc. This fully buffered board occupies any slot in the H8 and is dip-switch selectable to any port address on 8 byte boundaries. PSG X 4 for the H8 is priced at \$225, assembled only, with delivery anticipated by the time you read this.

These boards are of the finest quality using first run components, on double sided glass-epoxy stock with full solder-maks and silk-screening. Both make use of Schottky and MOS chips where possible, for the lower power consumption. An important feature (lacking in most designs) is an on-board crystal clocking oscillator, required for the most accurate reproduction of the notes of the equally tempered musical scale, although micro-tuning several notes between is possible. Both also include on board audio amp, and a monitor speaker, but may be configured for stereo with an external amp.

Prices include software on mini-disc to test your initializatiion, demo sound effects, sample games with sound, a program which allows you to play simple tunes from your keyboard, and more. Additional software packages will be made available in the future. Orders should include an additional \$5 for postage and handling, foreign \$8 extra.

H89 DOUBLE DENSITY HDOS FOR MAGNOLIA

From Dean Gibson of UltiMeth Corporation (the 400K drive software developer) 24025 Fernlake Dr, Harbor City, CA 90710, announces DKDDFLP.DVD, an HDOS 2.0 double density floppy disk device driver.

UltiMeth Corporation, in cooperation with Magnolia Microsystems, Inc, of Seattle, WA. Announces the availability (in September 1981) of disk device

driver software for HDOS 2.0 which supports Magnolia Microsystems' single-board, soft--sector, double density floppy disk controller for the Heath H89 and Zenity Z89 computer. The software, in conjunction with the disk controller board, will support the following disk drives under HDOS 2.0:

Four 8" (single/double sided, single/double density) each drive holding up to 1 million bytes; and Four 5 1/4" floppy drives (single/double sided, 40/80 tracks/side, single/double density), each drive holding up to .66 million bytes. (these are the Tandon TM-100-4 drives I have been handling--ed).

The Magnolia Microsystems disk controller board (with up to 8 of the above disk drives) may be used ALONG with the standard Heath mini-floppy disk controller board, (with up to 3 drives) for a total ON-LINE capacity under HDOS 2.0 of up to 8 million bytes!

It is intended that the 8" disk formats under HDOS will be compatilbe with Heath's 8" floppy (H-47).

The Magnolia Microsystems controller board, along with cables, CP/M version 2.23 (which supports both their disk controller board and the standard Heath mini-floppy disk controller board), and full documentation is available NOW for \$595 from Magnolia Microsystems, INC, 2812 Thorndyke Ave, West, Seattle, WA 98199 (206) 285-7266 or (800) 426-2841. Both 5" and 8" drives are also available from them (but you still won't find a better price for the Tandon's than H8SCOOP--ed). Add \$5 to all orders for handling and shipping US. (Last minute addition: My price on this board is \$510--ed)

This disk device driver to support the controller board under HDOS 2.0 will be available September for \$40 (which includes both 5 1/4" and 8" drivers, documentation, and shipping) from UltiMeth, the software being supplied on HDOS 5" single-sided 40TK hard-sector disk. Residents of above states add the appropriate sales tax. Oh yes, I assume this double density stuff will require SOFT SECTOR discs.

NEW! FROM HOYLE & HOYLE SOFTWARE--STEER KLEER

STEER KLEER is an obstacle course game for players from 4 to 104 years old. Four obstacle courses are provided, along with source code. It's a real time, absolute, variable speed, graphic, manual dexterity skill game. It's like eating nuts--bet you can't play just ONCE! Runs in 16K on H8 and H19 or H89 with any version HDOS. Price is \$10, or \$8 in orders of 5 or more. Of course, A REMARKABLE EXPERIENCE is still available for all brave explorers. Available from Hoyle & Hoyle Software, 716 S. Elam Ave, Greensboro, NC 27403.

FYI, \$25 first prize winner for completing "A REMARKABLE EXPERIENCE" is Don Harton, of Baltimore Maryland who sent the ending messages on June 23, 1981.

MORE ON IST SOFTWARE PRUF-REEDER

Burt Hulland sent me more info on the Software PRUF-READER I mentioned last month. It appears to have been written in Microsoft Fortran, but from the speed you would think it was in BASIC. It reads the text file at about 250 wpm. You can generate

or modify your own dictionaries. The price on this is \$39.95, and it's available from IST Software, Suite 409, 51 E. 42nd St., NY, NY 10017.

NEW SOFTWARE FROM HULLAND ENGINEERING

Burt has a new release of his modified operating system HDOS 0.1, called BH001.03. The changes consist mainly of documentation changes to make it a supplement to the HDOS 2.0 manual instead of the 1.6; however there are a few minor changes to improve compatibililty with 2.0. The price is still the same at \$40 by mail or \$59.95 at stores that carry it. For previous purchasers, \$10 plus the original disc will obtain it for you.

TURNKEY is another new item. It is a set of programs to convert any SYSGENed disk (any HDOS version, Burt's or Heath's) to operate a menu driven mode. Its purpose is to allow game discs, business systems, etc., to be made auto start for use by unskilled computer individuals. The menu and commands it generates are determined by the contents of a user supplied file, "MENU.FMT". Menu items, which can be called by numbers, single letters or words, can execute any system command or run any program. Input data can also be supplied to the program being run. The menu is automatically displayed when the disk is booted, and whenever any program exits to HDOS. There are also provisions for execution of system commands or programs that do not appear on the menu, and the entire TURNKEY system can be turned off or on at any time.

Burt says he thinks it will also work with the 8" drives, but hasn't tried it. The price of the TURNKEY system is \$19.95.

For those dealers or HUGS wishing to handle BURT's stuff, he's offering 40% off suggested price in quantities of 3 or more. This includes:

HDOS 0.1	\$59.95
Epson device driver	\$19.95
TURNKEY system mod	\$19.95

MORE FROM KEYBOARD STUDIO

The Keyboard Studio, 125 Aspen, Birmingham, MI 48009 (313) 645-5365 has added some new products to their growing line of software. THE PARALLELER, for \$24.95 allows you to parallel your console with another port, such as another console or a modem port. The idea is similar to DUALPORT which was mentioned June 1981 H8SCOOP, but is not supposed to be such a kluge. The PARALLELER, coupled with some add-on's (in progress by Tom Jorgenson) will allow you to operate your own bulletin board via HDOS. I should have a report on the entire bulletin board package in the next H8SCOOP. If you are interested in seeing this bulletin board operating, dial (314) 291-1850, which is the SLHUG. For \$18 you can have B-XREF which is a versatile MBASIC cross reference generator for HDOS and CP/M. The CP/M version is \$20. This program generates a complete cross-reference output of MBASIC keywords, user variables, or statement line numbers.

CTRL PATCH for \$18 allows you to use CTRL P to output what your screen sees to the printer. It works like CP/M CTRL P. There are additional programs and patches on this disc also. For a complete catalog listing, write THE KEYBOARD

STUDIO. (Last minute:--They now have an RS-232 interface voice synthesizer; introductory price of about \$350)

BETTER H-14 RIBBONS FROM PC&J GRAPHICS

The PC&J GRAPHICS CO. INC., POB 108, Deer Park, NY 11729 (516) 667-8076 is marketing a high quality nylon ribbon which fits the H-14 printer. It is "royal blue" in color and has terrific print characteristics, giving solid dark copy from the H14. They sell for \$4.95 each, or 3 for \$12 PPD.

SOFTWARE & HARDWARE FROM W.I.S.E

Wisconsin Intelligent Systems Engineering, Box 344-422 Third St., Baraboo, WI 53913 (608) 356-9432 has available for the H89 a 2S+2P+RTC board. This board contains 2 serial ports, 2 parallel ports, and a real time clock with battery backup. The PC board is double sided glass epoxy with plated through holes. There is a solder mask on both sides, and sockets for the 58167, the two 8250's, the two 1488's and the two 1489's. For those who have the standard serial board, the price is \$95, since you can take the UART'S and RS-232 chips from the Heath board. If you want all the parts, the cost is \$135, which does not include cables or battery. An on board NICAD battery goes for \$10. For more info on this, or additional option info, contact WISE.

They also have the OSBORNE business packages for the Heath/Zenith H89, which require CP/M and CBASIC 2, two drives, 48K and a 132 column printer. You can order manuals seperately for \$20 each. GENERAL LEGER is available complete for \$95, ACCOUNTS RECEIVABLE and ACCOUNTS PAYABLE for \$145, PAYROLL and PAYROLL COST ACCOUNTING for \$170, or the whole lot for \$375. These include the source object and data files. For more info, contact W.I.S.E.

NEW SKILL GAMES FROM EVRYWARE

EVRYWARE, POB 60802, Sunnyvale, CA 94088 has two games for the H19/H89 terminals--super graphics. INVASION puts you in command of earth's defenses. You control your missile launchers to shoot at an invasion force that is trying to land on earth. To make your job more difficult, the invasion drops wiggling bombs that explode on impact and turn into radioactive fallout whenever they are hit. The fate of earth is in your hands.

MISSILE CONTROL is a fast action game that will test your speed, accuracy and wits. You are under attack by a powerful enemy who has many long range guided missiles and bombers. You have the responsibility to protect six important cities from the onslaught of this menace. You do this by launching your own missiles which you can set to explode anywhere on the screen.

Both games are for the H89 or H8/H19 and require HDOS and 32K memory. Price is \$17.50 each. I got a chance to sit down with these for a few minutes to check them out. Very well done, with lots of action, and good fast moving graphics.

AUTO ALTER DISK DIRECTORY SIZE UTILITY

From John Efird of Palm Bay Software, 1109 SE Cypress Ln, Palm Bay, FL 32905 (305) 724-9092 comes DIRALT. This program will alter any HDOS disk so the file DIRECT.SYS only uses 4 sectors. It is

compatible with all known versions of HDOS, including 2.0. The program does not care if files already exist on the disk, as it will return the 14 sectors to the free space block.

The price on this utility is \$15 plus disk for the .ABS file, or \$20 plus disk for .ASM and .ABS file. Add \$7.50 if you want John to supply the disk.

8255 PPI CHIP INFO

For those folks who have the PC-12 clock-parallel port board by Don Long, Digi-Key Corp, Box 677, Three River Falls, MN 56701--is selling a book titled "microcomputer Interfacing with the 8255 PPI Chip". This book runs to 212 pages with a chapter each on mode 1 and 2. Several chapters on mode 0, covering simple I/O, bit set/reset, etc. At \$8.95 it is a good buy. Their catalog is free, and mainly covers IC's, components, some tools, and books of all kinds. This info was sent in by James Romoser, who says "I have placed several orders with them and never had any trouble. They are a 'Good Guy' in my book."

OFFERS

This month's offer came in at the last minute, so I can't tell you much about it. For \$8 order HVC disc, which is "From the Uni of Houston Victoria Campus". I was asked to supply the disc just as it came, no additions or deletions, so that's the way you'll get it.

There are some MNCH???? files for use with Munchkin from Walt Bilofsky's SOFTWARE TOOLWORKS. You'll have to have Munchkin to use these files. There are 8 in all, taking up 23 sectors. There is also the .DOC file, "WHATSUP.DOC" (23 sectors) with a description of the files on this disc. BIGFONT, which I feel is worth the price in itself, is an MBASIC program which prints large letters on an H89/H19 screen. What's amazing about this, is they are drawn just like a human hand would form the letters, rather than the usual top-to-bottom way a computer would do it. You can easily enter your own messages--really amazing.

FIBNIM is a nongraphic NIM program which allows many people to play at the same time. CAKE is an MBASIC program which is a super-simple computerized fantasy simulation for kids. It was written to show how to program a very simple CFS (computer fantasy simulation)

CHAIR (MBASIC) is also a CFS written to show how to implement CFS's. An earlier version was distributed by HUG (P/N 885-1088). This version is not graphic. These CAKE programs are by Furman Smith.

HAP by Furman stands for HACK and PACK, and what it does is hack off all comments from BHBASIC or MBASIC program, packs the program by removing all unnecessary blanks.

July/August RECREATIONAL COMPUTING, page 24 carried an article by Furman Smith on CFS programs. This disc, I feel, is an excellent value for everyone.

REQUESTS

REQUESTS FOR VCR BACKUP OF DISCS

There has been several requests for using a Video Cassette Recorder for backing up disks. This is commonly being done for hard disc systems, and the typical electronics package sells for about \$700. Since the bandwidth of VCR's makes them a natural for this purpose, it would be well worth looking into. If anybody has experience in the computer and Television field, this should be no major project. If anyone has any ideas, or a working VCR backup, contact H8SCOOP as many have been asking for this. The one that devises such a system for a low cost, could make lots of bucks. If you have the necessary interfacing experience, think about this project.

PAPER TAPE SOFTWARE WANTED

D. C. Shoemaker, 2000A Foxridge, Blacksburg, VA 24060 is looking for paper tape software to trade. Anyone interested contact the Major.

MEDIA CHECK PROBLEMS USING TRIONYX BOARD?

Gordon Richards, POB 7211A, Orlando, FL 32854 (305) 298-8729 wants to know if anyone has had any trouble running the media check of TEST17 using the Trionyx 64K M-H8 memory board in their computer. He says on his system the HDOS 2.0 media check worked fine using two Heath 8K boards, but when installing the Trionyx board the program bombed and the computer went dead. He says the check seemed to go almost to the end of the test before quitting. He gets the same results using HDOS 1.6, yet all other programs seem to run fine. Trionyx has not had any other complaints with this problem (and neither have I--ed). If anyone else has noticed this problem, please respond to H8SCOOP of TRIONYX.

TECHNICAL FORUM

HDOS INFO AND STUFF

Here's some information passed along to me by D. C. Shoemaker, from Al Heigl in Minnesota. I thought you might find it informative.

"JGL was named by, not for, Gordy Letwin (who now heads Softwin, Inc. -- Greg Chandler is doing HDOS at Heath now), and the switch's only function is to enable showing that C flag. That flag stands for only one thing -- Contiguous. In other words, a 'C'-flagged file has been written physically contiguously on the disk. Thus, when HDOS wants to read it, it doesn't have to figure out where the sectors are on the disk (via the GRT.SYS linkage table); it knows the file is flagged contiguous, it has looked up where it starts on the disk, and it has determined how long the file is -- so it can do one continuous read. This is designed to make the process as fast and smooth as possible. It happens that INIT and SYSGEN write certain system files contiguously, but others would be written that way, too, if anyone wanted to. (Also, on a freshly INITed or SYSGENed disk, most files added at first would happen to be written contiguously, just because the space is wide open then, but they are not forced contiguous, and thus are not flagged that way.)

"Now to Stand-Alone. First, it is thoroughly

debugged and commented well in the source code -- it's a simple function and there are no problems with it.

"When you type SET HDOS STAND-ALONE, no files at all are loaded into memory. It works the same as any other SET command; you are setting a bit flag in low memory, and setting a similar bit flag in the first sector of HDOS.SYS so that the memory bit will be set whenever that disk is booted up. (Vice versa for the NOSTAND-ALONE function).

"As for functioning, there are several aspects. You will note in the operations of PIP and Microsoft BASIC, it says that you may reset SY0: in either program; but if you do, an exit to HDOS will dismount all disks and tell you to reboot. This happens in this way because the Stand-Alone flag is not set, because it has been assumed that the system disk has been removed, and there's no telling what is on any disk that may be currently in SY0:. (It may well be the original, or a new system disk, but HDOS doesn't KNOW that -- hence, you must reboot).

"If, however, the Stand-Alone flag is set, HDOS will make one additional assumption -- that the user knows what he's doing! On any kind of "exit to HDOS", what the system does is load and transfer control to the system command interpreter SYSCMD.SYS. So, if you're running Stand-Alone, HDOS will over-ride its previous safeguard, and try to link to SYSCMD on whatever disk is in SY0: at the time. (And if it doesn't find it, it's Fatal System Error time!) Which means that after you have reset SY0: from PIP or MBASIC or whatever, when you exit to HDOS, you had better have SYSCMD.SYS on whatever disk you have in SY0:. In Stand-Alone, by the way, there's nothing wrong with initializing a data disk, copying SYSCMD.SYS onto it, and exiting to that after reset. Or making up a disk with PIP, SYSCMD, and BASIC, for example, and resetting to that once the operating system has been loaded into memory from a SYSGENed disk that you boot up on. Keep in mind that any device drivers that you might want to use must be LOADED into memory before SY0: is reset in any way, as any drivers not loaded in memory at reset time are removed from the list of what's available.

"The other aspects of Stand-Alone operation are in SYSCMD itself -- you may of necessity discovered them, although you didn't specify them in your article. Al is referring to the article on HDOS in February '81 Microcomputing written by D. C. Shoemaker--ed The command "RESET" now is a legal command, applicable to any drive (keeping in mind what disk you put in SY0: -- after a RESET, you are still running the SYSCMD program from the disk you reset from!) Also, you can type DISMOUNT SY0: without problems, but you'd better MOUNT something reasonable before you try to do much of anything at that point!.."

This is some useful information, and may help explain a few things to the more daring individuals or prompt some of you to experiment.

GOOD GUYS and BAD GUYS

Bruce Bennett writes that perhaps I should rate vendors as Good Guys, Mediocre Guys and Bad Guys. He writes "I noticed that one outfit was labeled a BAD GUY because one irate customer wrote you complaining about excessive delivery time. . ."

Although the point is well taken, I want to point out that I don't write up an outfit or person as GOOD or BAD on one notice from a reader. Even though I may only print the general context of one, I usually have several, which was the case in this instance. I also explain why they were so labeled, so the reader can infer what he wants. Being labeled a BAD GUY does not have to mean a hardened criminal, but does always mean unsatisfaction was expressed by several readers, and usually, to make matters worse, no great effort is taken to rectify the problem.

I realize people are human, and as thus are not perfect, and problems in shipping can arise. This does not bother me so much. What does bother me is when attempts are made by the person involved to trace down the problem, only to get little or no results. I have had this happen to me many times, and know what it's like, and the best thing I can think of is to let others know so they can avoid this same problem in the future.

Some good tips were sent in to me by Brian Hansen. "In the last few H8SCOOP's the BAD GUY reports seem to be increasing. I would like to also say the CPU SHOP in Charlestown Mass is giving me the run around about my order. I unfortunately missed the report in a previous H8SCOOP".

"I am enclosing a copy of a booklet sent to me by the FTC on a rule that mail order companies must follow. Maybe some of the information could be of some help to fellow H8SCOOP readers.

"I urge all readers to first try to resolve the problem with the company first since errors can occur, but if the run around starts, start complaining to every state, local and federal agency you can think of. If enough complaints are sent in on a company, action will be started, but no action will start without a pattern of complaints, so don't keep it to yourself, demand your rights!"

I must say I agree with Brian's philosophy. Too many of us talk and complain among each other, but that's exactly where it stops. If you have problems, DO SOMETHING about it, writing to H8SCOOP is a start, since we can at least keep our business away from them, and thus avoid the problem.

As far as Brian's problem with CPU Shop, he wrote in to tell me the CPU shop did answer his letter and explain the delay.

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SEPTEMBER issue deadline--August 13

Subscription Rates--\$15/Year (\$21 overseas AIRMAIL)--current issues
\$12/Year (\$18 overseas AIRMAIL)--first year back issues.

Complete set back issues from issue #1, April 1980 to #12 March 1981
still available for \$12 while supply lasts.

PHONE HOURS: Almost anytime you can catch me except sunset Friday to
sunset Saturday. NO COLLECT CALLS

Some points taken from the FTC report Brian sent me are as follows:

You must receive the merchandise when the seller says you will; . . .if you are not promised delivery within a certain time period, the seller must ship the merchandise to you no latter than 30 days after your order comes in; and . . .if you don't receive it shortly after that 30 day period, you can cancel your order and get your money back. The seller must also give you a free way to send back your answer (whether you want to wait longer or get your money back). If you cancel a prepaid order, the seller must mail you the refund within seven business days.

When I place a mail order, I always ask if the item is in stock, and am almost always told "yes" (and it hardly ever is!).

If you have a problem, you can contact your Better Business Bureau, they may be able to help. Contact the state or local consumer protection agency closest to the company. Call your local postmaster. Ask for the name and address of the appropriate postal inspector-in-charge. That person may also be able to resolve your dispute. Most important I feel is to contact the book, magazine, or newspaper publisher that carried the add. Publishers often try to resolve problems between their readers and advertisers, and may refuse further adds from that company if enough complaints are received. And keep in touch with GOOD GUYS and BAD GUYS through H8SCOOP, so you know who has been reliable in the past, and who has not.

I hope this space I have taken in this column this month, rather than point to GOOD and BAD guys will help you all in the future.

CLASSIFIEDS

FOR SALE--Static RAM bonanza. Three Heath 8K RAM memory boards with DIP switches for address, one Heath WH8-16 16K RAM memory board, all manuals etc., included, Seven TMS 4044-4 4K RAM memory chips. I pay shipping, \$419 takes all. Brian Branson (see below).

FOR SALE--One HP 2621A video display terminal in original box. This is a professional unit with 2 pages of memory, scroll up/down, previous/next page, addressing, editing, N-key rollover, auto repeat any key, detached keyboard, capable of displaying control characters as a selectable mode. This is an extremely nice terminal in mint condition. I pay shipping. \$1095 takes it. Brian Branson 2255 Cahuilla Rd #108, Colton, CA 92324, phone (714) 824-0144.